

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 16

(1) Research Program/Project/Study Title: Development and Evaluation of Senior High School Core Mathematics Subjects' Achievement Test	
Proponent Name:	Alvin D. Tenorio
Academic Rank:	Assistant Professor IV
Co-Authors (with academic rank):	Emir Lenard S.F. Sicangco Associate Professor II
Campus/Satellite College/Address:	CapSU Pontevedra Campus – Bailan, Pontevedra, Capiz Tarlac State University Lucinda Campus – Tarlac City
Telephone/Cell Phone/Fax Number:	0956-690-7315
E-mail Address:	adtenorio@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
Municipality/City:		Tarlac City							
Province:		Tarlac							
Region:		Region III							
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Education		(8) Discipline/Field of Specialization Educational Measurement, Mathematics Education							

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	
	Document Title:	Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 2 of 16

(9A) Abstract

This study aimed to develop a Senior High School Core Mathematics Subjects' Achievement Test (SHSCMSAT) to measure students' achievement in senior high school core mathematics subject. Using descriptive research, 150 items were developed initially, submitted to domain and technical experts, and piloted to 337 senior high school students of a public national high school. Item analysis which includes computation of item difficulty and discrimination indices as well as analysis of options were performed. Validity was established by acquiring content, criterion, and construct-related evidences. The KR 20 coefficient of internal consistency was computed to establish the reliability of the instrument. Results of the analysis showed that the final 50-item SHSCMSAT could parsimoniously gauge students' understanding of key concepts in senior high school core mathematics subjects. Items of the test were of intermediate difficulty and could fairly to excellently discriminate good and poor students. Further, the distractors were equally effective and appealing to students who have not mastered the lesson but the correct answer was apparent from the choices on the part of students who really learned the lesson. The final version of the test had an acceptable reliability estimated by KR 20 coefficient of .778 and is considered valid in measuring students' achievement in senior high school core mathematics subjects as supported by content, criterion, and construct-related evidences collected. Suggestions for subsequent trial run of the instrument, and recommendations for use of the SHSCMSAT are discussed.

(9B) Keywords

Test Development, Educational Measurement, Achievement Test, General Mathematics, Statistics & Probability, Classical Test Theory

(10) Introduction

The enactment of Republic Act No. 10533 also known as the Enhanced Basic Education Act of 2013 has made significant changes in the Philippine basic education system. As stipulated in Sec. 2 of the said law, the rationale behind this major revamp in the educational system is to provide Filipino learners with functional basic education which will develop them into productive and responsible citizens equipped with the essential competencies, skills and values for both life-long learning and employment (Enhanced Basic Education Act, 2013). One of the major changes is the addition of two years in the third stage of compulsory basic education which is tagged by the law as the senior high school. The senior high school is the last two years of secondary education wherein the students are made to choose a track based on their interest and life goals. The tracks include academic, technical-vocational livelihood, sports, and arts and design. These additional two years in secondary education aim to provide students with a good grounding on core and general education subjects in preparation for college, equip them with necessary skills should they decide to embark in the world of work, and among others.

One of the core subjects that the senior high school students need to take regardless of the track they enroll is mathematics. The core mathematics subjects required among senior high school students are General Mathematics and Statistics and Probability. General Mathematics is offered in the first semester of the 11th grade followed by Statistics and Probability in the second semester. The sequencing of mathematics subjects in the curriculum is by spiral progression approach to ensure mastery of knowledge and skills after each level. In addition, mathematics subjects were harmonized with that of the tertiary education curricula to ensure readiness in college and avoid duplication of basic education subjects.

The aims of mathematics in basic education is to develop the critical and analytical thinking skills of students which are necessary for them to become informed citizens and as they pursue higher education and compete in a technologically oriented workforce (SEI-DOST & MATHTED, 2011). Mathematics education has long been recognized to be playing an important role in national development. This claim serves as an impetus for experts to put forward innovations in the curriculum for the advancement of mathematical literacy of Filipino students. Results of evaluations released by international organizations regarding mathematical achievements of Filipino students also posed a challenge to improve the curriculum to keep up with the international standards. UNESCO, in their report about the challenges in basic mathematics education highlighted that both national and international evaluations show that, on completion of basic education, many pupils' mathematics

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	
	Document Title:	Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 3 of 16

knowledge and competencies fall short of the expected level (UNESCO, 2012). In addition, the results of the Trends in Mathematics and Science Study (TIMSS) in 2003, revealed that Filipino Grade 8 students who took the mathematics test scored 378 and ranked 34th of 38 countries (National Center for Education Statistics [NCES], n.d.). While on the latest TIMSS-Advanced conducted in 2008, the Philippines ranked 10th out of the 10 countries who participated with an average scale score of 355 (Ogena et al., 2010). These statistics are reflective of the low performance in mathematics of Filipino students compared to their international peers.

The goal to increase mathematical literacy among Filipino students remains far from reach. Education specialists hope that the reformed basic education curriculum could somehow improve the performance of Filipino students in mathematics. But to gauge students' mathematical learnings, a carefully constructed assessment tool is necessary. This means that the assessment tool can measure the expected learning outcomes among students and has undergone the stringent process of test development. The Department of Education (DepEd) through DO No. 55, s. 2016 known as the Policy Guidelines of the National Assessment of Student Learning for the K to 12 Basic Education Program ordered the conduct of exit assessment to determine if learners are meeting the learning standards of the Elementary, Junior High School, and Senior High School curriculum (DepEd, 2016). For senior high school, the exit assessment as specified in the guidelines shall cover 21st century skills and the core Senior High School learning areas of Languages, Humanities, Communication, Mathematics, Science, Social Science, and Philosophy. Considering the number of subjects that the exit assessment shall cover, it is seen that some learning competencies in math subjects will be left out. For a math teacher who wants to find out student's achievements in the specific topics under the core mathematics subjects, the result of exit assessment can't fully give the needed information owing to its non-domain-specific nature. As of this writing, and to the best knowledge of the researchers, no achievement test has been published yet specifically to assess senior high school students' achievement in their core mathematics subjects. The need for the said achievement test is also felt by graduate students who plan to undertake researches on mathematics achievement of senior high school students.

The above-mentioned gaps inspired the authors to develop an achievement test in senior high school mathematics subjects. It is seen that the development of the achievement test is beneficial to mathematics teachers in determining students' strengths and weaknesses in the subject as well as in identifying those who have learning difficulties or those who are likely to develop such problems. Also, the instrument is deemed helpful in assessing students' achievement in mathematics which could provide valuable inputs to curriculum experts and mathematics specialist to further improve the quality of mathematics teaching in the basic education. Finally, the development of this achievement test could benefit future researchers who would like to use the instrument in their research and those who would like to develop similar instrument to assess students' learning in mathematics.

Generally, this study aimed to develop and evaluate a senior high school core mathematics subjects' achievement test (SHSCMSAT) following the standards of the test development process. Specifically, answers to the following question will be sought.

1. What are the difficulty and discrimination indices of the items of the achievement test?
2. What content, criterion, and construct-related evidences can be acquired to establish the validity of the achievement test?
3. What is the internal consistency reliability of the achievement test?

(11) Theoretical Framework

Classical Test Theory (CTT) is a psychometric framework used to understand and evaluate the characteristics and quality of psychological tests and measurement instruments (Urbina, 2014). The theoretical framework of CTT revolves around three main concepts: observed score, true score, and measurement error.

1. **Observed Score:** The observed score is the score obtained by an individual on a test or measurement instrument. It represents the numerical value or response that is recorded or observed. It is the sum of the true score and measurement error.

2. **True Score:** The true score is the hypothetical or latent score that reflects the individual's actual standing on the construct being measured. It represents the underlying attribute or trait that the test intends to measure, free from any measurement error. However, since it is not directly measurable, it is estimated indirectly through the observed score.

3. **Measurement Error:** Measurement error refers to the variability or inconsistency between the

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	
	Document Title:	Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 4 of 16

observed score and the true score. It represents the factors that cause the observed score to deviate from the true score. Measurement error can arise from various sources such as test administration conditions, scoring errors, or individual-specific factors.

The central premise of CTT is that the observed score is a combination of the true score and measurement error. CTT provides a framework to estimate and separate these components, allowing researchers to evaluate the reliability and validity of a test. The reliability of a test refers to its consistency or stability, while validity refers to the extent to which the test measures what it intends to measure.

Key concepts within CTT include:

1. Reliability: Reliability is the extent to which a test produces consistent and stable results. CTT provides various reliability coefficients, such as the Cronbach's alpha, test-retest reliability, and inter-rater reliability, to assess the consistency of scores over time, across different items, or between different raters.

2. Validity: Validity refers to the extent to which a test measures the construct or trait it intends to measure. CTT does not provide specific validity measures but offers a framework to evaluate different types of validity evidence, such as content validity, criterion validity, and construct validity. Validity evidence can be gathered through empirical studies, expert judgments, or theoretical arguments.

3. Item Analysis: CTT includes methods for item analysis to evaluate the quality and characteristics of individual test items. Item analysis examines item difficulty (the proportion of individuals who answer an item correctly), item discrimination (the extent to which an item distinguishes between high and low scorers), and item-total correlations (the relationship between an item and the overall test score).

Overall, CTT provides a theoretical framework for evaluating the quality, reliability, and validity of psychological tests. It helps researchers and practitioners understand the psychometric properties of tests and make informed decisions about their use in assessing individuals' attributes, abilities, or traits.

(12) Methodology

Research Design

The test development was descriptive in nature. It commenced with the identification and specification of the purpose of the instrument to be developed. Literature review was done to have an in-depth understanding of the goal of the senior high school mathematics curriculum. Documents including department orders, curriculum guides, and textbooks were perused to ascertain the competencies to measure in the test. To ensure that the resulting test will be psychometrically sound, the researchers observed the stringent procedures and standards of test development.

Pilot Testing Respondents and Study Site

The pilot testing was done in one of the national high schools in Region III. The said school is one of the biggest public high schools in the said region and holds the distinction of being the seat of the oldest public high school system in the Philippines. It serves more than 7,000 students from Grade 7 to 12. The national high school which served as the study site offered the four major tracks in their senior high school namely: Academic, Technical-Vocational Livelihood (TVL), Sports, and Arts and Design. Under the Academic Track, the strands offered were Science, Technology, Engineering, and Mathematics (STEM), Accountancy and Business Management (ABM), Humanities and Social Sciences (HUMMS), and General Academic Strand (GAS). The TVL strands offered were Electrical Installation and Maintenance (EIM), Information and Communication Technology (ICT), and Home Economics (HE).

Originally, 358 randomly selected senior high school students participated in the pilot testing. But after reviewing the returned answer sheets, 21 students were discarded from the analysis either because of incomplete or lost answer sheets. The final number of students included for analysis was 337. Of the 337 students, 157 (46.59%) of them were males and 180 (53.41%) were females. The respondents' mean age was 17.57 years, typical for a senior high school student. Out of the 337, 34 (10.01%) were enrolled in STEM, 134 (39.76%) in HUMMS, 122 (36.20%) in TVL, and 47 (13.95%) were enrolled in the ABM strand. When type of junior high school attended was considered, only six of the 337 respondents graduated in a private school while the rest finished their junior high school in public school.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH		Revision No.	00
				Effective Date	
				Page	Page 5 of 16

Test Description

The achievement test was developed to assess students' achievement in senior high school core mathematics subjects. The test was a multiple choice type initially composed of 150 items covering the two core mathematics subjects in senior high school namely General Mathematics and Probability and Statistics. General Mathematics covered three broad lessons on Function and their Graphs, Basic Business Mathematics, and Logic. Statistics and Probability covered lessons on Random Variables and Probability Distributions, Normal Distribution, Sampling and Sampling Distributions, Estimation of Parameters, Tests of Hypothesis, and Correlation and Regression Analyses. The items in the test ought to measure students' achievement of the content and performance standards in each topic covered by the two core subjects.

Test Development Procedures

After the purpose of the test development had been laid-out, the table of specification (TOS) was prepared which served as the blueprint in the construction of the test items. The TOS contained the topics and competencies to be assessed, the level of thinking skills to be measured, and the placement of items which will measure the learning competencies. The learning guide provided by the DepED served as basis of the topics covered by the test. The learning guide contained information on the subject's content and performance standards and learning competencies expected of the students to master at the end of instruction. Each competency has corresponding one item in the test to ensure that all learning competencies were covered in the test. The behavioral objectives in measurable terms stated in the learning competencies were made as bases of the level of thinking skills measured in each item. Four textbooks used by senior high school students including the target respondents served as references in making the items to ensure that the terms and symbols used in the test were the ones used by the target respondents in their class. Other books and online resources were also utilized to supplement the textbooks.

The two authors formulated the items in the test separately. One constructed the items in General Mathematics and the other constructed questions in Statistics and Probability. After the construction of the items, it was submitted for review to four domain experts who were teachers teaching the subject and one technical expert in test development. Comments of domain and technical experts were incorporated to come up with the final draft of the test. After finalizing the contents of the instrument and the answer sheet, it was pilot tested to the identified school. The researchers, with the help of the subject teachers of the respondents administered the test. The respondents took the General Mathematics first followed by the Statistics and Probability. The answer sheet of the respondents were scanned using the Zipgrade App and exported to excel for analysis.

Item Analysis Using Indices of Difficulty and Discrimination, Validity, and Reliability

Item analysis is a method of reviewing items on a test, both qualitatively and statistically, to ensure that they all meet minimum quality-control criteria (Varma, n.d.). Qualitative item analysis requires the expertise of people who are considered content and technical experts. This analysis is done to ensure that the items of the test meet the minimum criteria and whenever quantitative data are not yet available for statistical analysis. On the other hand, statistical method in item analysis uses real-world quantitative data after the test has been pilot tested to the target respondents. Several test statistics were computed with the aim of identifying problematic items which have slipped through the content and technical experts during the qualitative item analysis phase. Both methods of item analysis aim to improve test items (including the options in the case of selection types) to ensure that whatever results that may be generated by the test are indicative of the students' ability and mastery of the learning content.

In statistical method of item analysis, two indices are generated to determine the level of difficulty of the items as well as to find out if it functions well in discriminating students who have really learned the lesson from those who only guessed during the exam. These are referred to as the difficulty and discrimination indices. Before these indices are calculated, students are arranged according to the total scores they obtained from the exam. Those students whose scores belong to the upper 27% are considered to be in the upper group while those with scores equivalent to 27th percentile or below are considered to be in the lower group. The difficulty index is obtained by getting the average of the proportion of students who got the correct answers from the lower and upper group. Difficulty index can range from 0 to 1 with 0 indicating a very difficult item and 1 a very easy item. Discrimination index, on

	Document Type:	DOCUMENTED INFORMATION	
		ISO 9001:2015	
	Document Title:	COMPLETED RESEARCH	
		Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 6 of 16

the other hand, is computed by getting the difference of the proportion of students who got the correct answer between the upper group and the lower group. The discrimination index can range from -1 to 1 where negative values indicate that a lot of students in the lower group got a correct answer than those from the upper group and positive values otherwise. Most literatures suggest that for an item to discriminate effectively, it must have a discrimination index of at least 0.40.

Another important concept in instrument development is reliability. Reliability refers to the consistency of the scores obtained – how consistent they are for each individual from one administration of an instrument to another and from one set of items to another (Fraenkel et al., 2022). Urbina (2014) said that reliability coefficients provide test users with some information concerning the magnitude of error that is likely to enter into scores from various sources. The four major sources of errors in assessment are content sampling error, occasion sampling error, examiner error, and scorer error. Reliability can be investigated using different procedures. The first group of procedures requires examinees either to take a test twice or to take two forms of the test. The first group includes test-retest, test-retest with different scorers, test-retest with different examiners, parallel-form-immediate, parallel-form-immediate with different scorers, parallel-form-with-interval, parallel-form-with-interval and different scorers, parallel-form-with-interval and different examiners, parallel-form-with-interval and different examiners/scorers (Anastasi & Urbina, 1997). The first group of procedures poses difficulties logistically making the second group more appealing to test developers. The second group of procedures requires only one form of test and single administration. This includes split-half, Kuder-Richardson 20, and Coefficient Alpha collectively known as internal consistency estimates. These procedures are based on the internal relationships among parts of the test. The two groups of procedures take into account different sources of error. However, Urbina (2014) gave a caveat that in evaluating reliability data, one must keep in mind the fact that these estimates are affected by the characteristics of the sample for which they were computed and may or may not generalize to other groups of test takers.

Aside from reliability, validity should also be of paramount concern in the process of test development. Validity has been defined in 1921 by the National Association of the Directors of Educational Research as the extent to which a test measures what it purports to measure (Rogers, 1995). This classical definition of validity was ratified by several testing experts because it was seen as problematic from the perspective of current testing standards. Validity, in the current testing standards, is defined as the degree to which evidence and theory support the interpretation of test scores for proposed uses of tests (AERA, APA, & NCME, 2014). Also, validity is now considered as a unitary concept and what used to be labelled as its types are now viewed as convenient categories for accumulating evidences to support the validity of an interpretation. Keith Waugh & Gronlund (2013) listed and described four types of evidences to look into when evaluating the validity of a test namely content-related, criterion-related, construct-related, and consequences. Content-related evidences provide information about how adequate the sample of assessment tasks represent the domain of task to be measured. Criterion-related evidences support claim that performance on the assessment predicts future performance (predictive study) or estimate present performance (concurrent study) on some other valued measure called a criterion. On the other hand, construct-related evidences answer the question “how well can performance on the assessment be explained in terms of psychological characteristics?” While consequences as evidence of validity justifies how well the use of assessment serves the intended purpose (e.g., improve performance) and avoid adverse effects (e.g., poor study habits) (Keith Waugh & Gronlund, 2013).

Validity of the instrument can be investigated qualitatively and statistically. One of the statistical approaches in investigating validity is through factor analysis. The principal goals of factor analysis are (a) to reduce the number of dimensions needed to describe data derived from a set of measured variables, and (b) to investigate the structure that accounts for the interrelationships between the variables, in order to classify them and better understand the nature of the information that they provide (Urbina, 2014). There are two basic ways to conduct factor analysis. First is exploratory factor analysis or EFA. EFA explores the data and provides the researcher with information about how many factors are needed to best represent the data (Hair et al., 2014). This means that factors are derived statistically and not dictated by existing theories. The second way is by confirmatory factor analysis (CFA). CFA, as opposed to EFA, works in a way to test hypotheses, or to confirm theories about factors that are already presumed to exist. CFA statistics tell how well theoretical specification of the factors matches reality (the actual data) (Hair et al., 2014).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH		Revision No.	00
				Effective Date	
				Page	Page 7 of 16

(13) Results and Discussions

Initial Pool of Items

The test was initially composed of 150 items, half of which came from General Mathematics and the other half from Statistics and Probability. Of the 75 items in General Mathematics, five were intended for the topic on Functions and their Graphs, nine for Rational Functions, 26 for Inverse Functions, Exponential Functions, and Logarithmic Functions, 11 for Simple and Compound Interests, and Simple and General Annuities, six for Stocks and Bonds, four for Business and Consumer Loans, 11 for Propositional Logic, Syllogisms and Fallacies, and three covered topic on Methods of Proof and Disproof. On the other hand, out of the 75 items in Statistics and Probability, 10 tested students' learnings about Random Variables and Probability Distribution, six on Normal Distribution, eight on Sampling and Sampling Distributions, 22 on Estimation of Parameters, 17 on Test of Hypothesis, and 12 on Correlation and Regression Analyses.

Item Analysis Using Indices of Difficulty and Discrimination and Option Analysis

The first item analysis done was the computation of indices of difficulty and discrimination. The scores of 337 students in the 150-item test were arranged in ascending order to identify those who belong in the lower and the upper group. Ninety one students whose total scores belong to the upper 27% were included in the upper group, while 91 students who got a score belonging to the lower 27% were put in the lower group. The remaining 155 students whose scores fall between the upper and the lower 27% were excluded in the analysis.

The difficulty level of the item was computed by getting the average of the proportion of students who got the correct answers from the lower and upper group. The item with computed difficulty index of 0.01 – 0.14 was interpreted as very difficult, 0.15 – 0.39 as difficult, 0.40 – 0.70 as desirable, 0.71 – 0.85 as easy, and 0.86 – 1.00 as very easy item (Wiersma & Jurs, 1990).

The discrimination index of the item was computed by getting the difference of the proportion of students who got the correct answer between the upper group and the lower group. The item whose computed discrimination index was 0.19 and below was interpreted as a poor item, 0.20 – 0.29 as fair, 0.30 – 0.39 as good, and 0.40 and above as excellent item (Wiersma & Jurs, 1990). Table 1 shows the distribution of the 150 items according to their difficulty and discrimination index.

As seen in the table, a lot of items obtained a poor discrimination index and were considered difficult while the rest scattered under excellent and fair discrimination index and desirable to difficult difficulty index. Items whose difficulty index was interpreted as desirable or difficult and whose discrimination index was interpreted as either fair, good, or excellent were considered in the selection for the final version of the test. As reflected in the table, only 46 items were able to meet the specified conditions in terms of the item difficulty & discrimination index. Since the target number of items for the final version of the test was 50, the researchers deliberated and considered the inclusion of four items with difficulty and discrimination index marginal to the desired coefficients. Of the final 50 items, 27 came from the subject General Mathematics while the remaining 23 items came from Statistics and Probability. Table 2 shows the 50 items that were considered for the final version with their corresponding difficulty and discrimination indices.

Aside from the analysis of the items in terms of their difficulty and discrimination indices, an option analysis was also performed to determine the plausibility of the distractors. The plausibility of the distractors was determined by getting the proportion of students who chose the options in each item both from lower and upper group. Table 3 shows the truncated results of the option analysis done. The complete results of the option analysis will be available upon request to the researchers.

As seen in the table, options in item 1 were effective considering that eight out of 10 students in the upper group selected the correct answer while students in the lower group were spread out in the four choices. For item 2, though more than half of the students in the upper group chose the correct option, still four out of 10 of them got the item wrong. The options though worked well for the students in the lower group. Finally, options in item 3 were effective for students in the upper group since majority of them got the item right while its somewhat effective for students in the lower group. Option D in this item however was not effective since few students from the lower group selected it.

The 46 items that met the specified conditions in terms of the item difficulty and discrimination, together with the four items considered essential by the experts, composed the 50 items subjected to factor analysis in the process of acquiring construct-related evidence of validity.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	
	Document Title:	Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 8 of 16

Analyses of Validity

As discussed earlier, validity, in the lens of current testing standards, hinges on the evidences we can provide to undergird any interpretations and inferences to be made on the basis of test results. The sources of evidences to support validity of the test results are content-related, criterion-related, construct-related, and consequences-related. For the analysis of validity of the test results, the researchers considered the first three sources of evidences. The discussions that follow detail how the evidences were acquired to support the validity of the test results.

Content-Related Evidences of Validity

To provide content-related evidence of validity is to show that the test samples adequately the important task and competencies consistent with the defined objectives of the subject. To ensure that the test covers the essential task and competencies expected of the students, the TOS was prepared. The topics, competencies, and level of thinking skills specified in the TOS were based on the learning guide provided by the DepEd. The behavioral objectives in measurable terms stated in the learning competencies were made as bases of the level of thinking skills measured in each item.

After preparing the test items with the TOS as blueprint, a copy of it was given to four domain experts for review. The domain experts were teachers teaching the subjects in senior high school. The researchers asked the domain experts to examine the overall test if it sampled adequately the topics in the subject and let them scrutinize the test items in terms of its level of difficulty, appropriateness of the thinking skills measured, and alignment to the specified thinking skills.

Comments of the domain experts were minimal and mostly on the level of thinking skills measured by the test items and the options. Some items that were originally under analysis were reclassified to understanding and vice versa. As to the options, domain experts modified some to make it plausible based on their experiences as to the common mistakes of students in a multiple choice type of test. The researchers deliberated the comments and suggestions of the domain experts and incorporated those which were deemed appropriate to come up with the final version of the test for the pilot testing.

Criterion-Related Evidence of Validity

As described by Fraenkel et al. (2022), criterion-related evidence of validity refers to the relationship between scores obtained using the instrument and scores obtained using one or more other instruments or measures (often called a *criterion*). Criterion-related evidence of validity is typically expressed as correlation coefficient – the correlation between the test (predictor) score and the criterion score (Allen & Yen, 1979). There are two ways to estimate correlation between test score and the criterion namely predictive and concurrent. Predictive validity involves using test score to predict future outcomes while concurrent validity uses test score and correlates it with the criterion scores obtained at the same time.

For the criterion-related evidence of validity, the researchers computed for the concurrent-validity coefficient by correlating the average grade of the students in General Mathematics and Probability and Statistics to their scores in the achievement test given. The test of correlation was done twice, first with scores of the students in all the 150 items and the second with their scores in the 50 items. Spearman's rho was used to test the correlation between the two variables because the assumption on normality was not met. Results of the test for correlation are presented in Table 4.

From the table, it can be surmised that there was a correlation between the achievement test scores and grades in senior high school core mathematics subjects both for 150 and final 50 items. This shows evidence of concurrent validity of the test or it is correlated with the criterion scores taken at the same time. It can be noticed however that the correlation value for 50 items was greater than that of the 150 items implying that the 50-item test can obtain scores that would correlate better with the criterion score. Correlation results further provided evidence that those who obtained high scores in the achievement test were also those students who performed well in their senior high school core mathematics subjects.

Construct-Related Evidence of Validity

Cronbach (as cited in Urbina, 2014) suggested the use of the term construct validity to designate the *nomological net*, or network of interrelationships between and among theoretical and observable elements that support a construct. Allen & Yen (1979) further described construct validity of the test as

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	
	Document Title:	Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 9 of 16

the degree to which it measures the theoretical construct or trait that it was designed to measure.

To provide evidence of construct validity of the test, the final 50 items were subjected to a factor analysis to show that the test really measures the two constructs which were the General Mathematics and Statistics and Probability. Confirmatory factor analysis was done in SPSS with Principal Component Analysis as method of factor extraction and Varimax as method of factor rotation. The data were found to be factorable since the Kaiser-Meyer-Olkin measure of sampling adequacy coefficient obtained was .66 and the Bartlett's Test of Sphericity was significant indicating that the correlation matrix of the items was not an identity matrix. The factor loading threshold was set by the researchers to .10. The result of the confirmatory factor analysis is reflected in Table 5.

As seen in the table, 28 items loaded in factor 1 while 21 items loaded in factor 2. The items which loaded in factor 1 were items 25, 10, 2, 11, 21, 33, 4, 87, 36, 103, 95, 3, 24, 15, 77, 45, 100, 1, 5, 71, 106, 91, 129, 28, 141, 64, 32, and 114. On the other hand, items which loaded in factor 2 were items 58, 98, 60, 63, 92, 43, 73, 142, 88, 38, 57, 105, 128, 123, 120, 178, 27, 96, 97, 104, and 75. Twenty-one items crossloaded in the two factors but they were placed in the factor which their loading is high. In addition, these items will be analyzed and improved in the future test run. On the other hand, item 118 did not load in any of the two factors but this item will not be discarded since it is an important item in the test as judged by the content experts. The researchers, however, will improve the stem of the said item and its options in the final version of the achievement test. Examination of items which loaded in factor one showed that 19 out of 28 were originally items in the General mathematics and scrutiny of items in factor two revealed that 13 out of 21 were originally items in Statistics and Probability.

Reliability Analysis

Reliability as described earlier is based on the consistency and precision of the results of the measurement process (Urbina, 2014). Reliability coefficient is computed to approximate measurement errors related to the process of administering the test, nature of the test taker, or the quality of the instrument used.

In the development of this achievement test, the internal consistency reliability was computed. This approach in determining the reliability of the instrument was chosen because of logistic considerations. Specifically, KR 20 internal consistency estimate was used because the items were dichotomously scored. Using the SPSS, the KR 20 coefficient of internal consistency was computed for both 150 and 50 items. The Cronbach Alpha model was chosen in the SPSS but the results were interpreted as KR 20 coefficients of internal consistency. Results of the reliability analysis are shown in Table 6.

From the table, it can be gleaned that the internal consistency of the achievement test improved when the total number of items was reduced to 50. Items deleted from the reliability analysis were those items found to be problematic during the item analysis. The KR 20 coefficient of the final 50-item test was .778 implying that the reliability of the scores obtained by the instrument was acceptable.

The internal consistency reliability of the items belonging to the two factors were also computed to find out if the items in each factor hold together or really measure the same construct. Results in Table 7 showed that items in factor 1 have an internal consistency reliability coefficient of .743 and interpreted as acceptable while those under factor 2 was .611. Though internal consistency of items in factor two was low, taken as whole, the 50-item achievement test still has an acceptable internal consistency reliability.

(14) Conclusions and Recommendations

Conclusions

Since test results are crucial in making decisions about individual students, it is only imperative that a sample of students' learning be obtained using a carefully constructed test. This study, hence, was an attempt to develop and evaluate an achievement test that could measure students' learning in senior high school core mathematics subjects.

This research has come up with a 50-item test that could parsimoniously gauge students' understanding of key concepts on senior high school core mathematics subjects. Items of the tests are of intermediate difficulty and could fairly to excellently discriminate good and poor students. The test, therefore, is suitable to students of varying ability. Moreover, distractors are equally effective and appealing to students who have not mastered the lesson but the correct answer is apparent from the

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH		Revision No.	00
				Effective Date	
				Page	Page 10 of 16

choices on the part of students who really learned the lesson.

The test developed is deemed valid as supported by evidences collected. Evidence based on content is clearly demonstrated by strictly following the curriculum guide provided by DepEd. Agreement among domain and technical experts further upholds the test's content-related evidence of validity. This is an indication that items in the test collectively sample content domains, cognitive processes, skills, and types of performance expected among students after taking the course. Criterion-related evidence showing that correlation exists between the achievement test scores and grades in senior high school core mathematic subjects implies that an inference can be validly made that when a student obtained a high score in the achievement test, then it could be concluded that he or she is also performing well in the subject and vice versa. This claim can be defended on logical grounds that the test scores are really reflections of students' performance and not by chance or product of guessing. In addition, construct-related evidence of validity as demonstrated by performing factor analysis shows two underlying constructs in the test. This confirms the two structures that account for interrelationship among items in the test. This result suggests that the test is really measuring constructs of the two subjects specified and not any other constructs not covered by the subject or by the goal of developing the test itself.

Lastly, internal-consistency reliability of the test is acceptable and thus provides evidence of cohesiveness of the items in the test. Deducing from the result, it can be concluded that the achievement test is unidimensional and is measuring students' competence in senior high school core mathematics subjects namely General Mathematics and Statistics and Probability.

Recommendations

First and foremost, the nature and the number of respondents may be considered in piloting the test. In the current study, the number of students coming from the different strands is not proportional and there was even no student who was from the General Academic Strand (GAS), Sports, and Arts and Design. Thus, a bigger sample proportionally composed of students from all strands may be taken into consideration. This would further ensure that the test will be suited to all types of students including those who are not really academically inclined. This suggestion is also seen as something that may improve the results of factor analysis specifically in increasing the factor loading coefficient of the items. A mix of students coming from both public and private senior high schools may also be considered.

The testing environment and time may also be well-prepared and planned. As much as possible, external factors inherent in the environment such as the condition of the room, noise, and other aspects be fixed before administering the test to minimize its impact to the test takers. Moreover, ample time may be given to answer the test to avoid fatigue among test takers.

As to the test itself, future researchers may consider the test items actually used by subject teachers in their exams and observe if it would function really well when used in an achievement test. Moreover, the items including the options that did not survive during this first trial run may be improved for the second administration. Examination of items with multiple loadings in the result is also suggested.

As to the suggestion in acquiring evidence of validity, the researchers recommend that future studies consider collecting evidence as to the predictive validity of the instrument. In addition, other methods in estimating the coefficient of reliability is recommended like test-retest or alternate form to estimate content and occasion sampling errors.

The researchers further recommend qualitative interview of the teachers and students to find out other possible sources of errors, problems regarding the instrument, and other factors which could help improve the subsequent administration of the test.

Lastly, even with some limitations identified, the researchers recommend the use of the SHSCMSAT to senior high school teachers and researchers who want to measure the achievement of senior high school students in core mathematics subjects.

(15) Output

The major output of this study is the 50-item Senior High School Core Mathematics Subjects' Achievement Test (SHSCMSAT) that can be used by senior high school teachers or researchers who would like to determine the achievement of senior high school students in senior high school core mathematics subjects.

(16) Technology Transfer Activity Conducted

Not applicable for this study.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 11 of 16

(17) Economic Analysis

Not applicable for this study.

(18) Acknowledgments

The researchers would like to thank the principal of the national high school where the study was conducted for allowing them to administer their instrument to the senior high school students.

(19) References

- Alen, M., & Yen, W. (1979). *Introduction to measurement theory*. Brooks/Cole Publishing Company
- Anastasi, A., & Urbina, S. (1997). *Psychological testing* (7th ed.). Prentice Hall.
- American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. American Educational Research Association.
- Department of Education. (2016). *Policy guidelines on the national assessment of student learning for the K to 12 basic education program* (Order no. 55). Author.
- Enhanced Basic Education Act of 2013. (2013, May 15).
https://www.senate.gov.ph/republic_acts/ra%2010533.pdf
- Fraenkel, J. Wallen, N., & Hyun, H. (2022). *How to design and evaluate research in education* (11th ed.). McGraw-Hill Companies.
- Hair, J., Black, W., Babin, B., & Anderson, R. (2014). *Multivariate Data Analysis* (7th ed.). Pearson Education Limited.
- Keith Waugh, C. & Gronlund, N. (2013). *Assessment of Student Achievement*. (10th ed.). Pearson Education, Inc.
- National Center for Education Statistics. (n.d.). *Trends in international mathematics and science study*.
<https://nces.ed.gov/timss/countries.asp>
- Ogena, E., Lana, R., & Sasota, R. (2010). *Performance of Philippine high schools with special curriculum in the 2008 trends in international mathematics and science study (TIMSS-Advanced)*. Paper presented at 11th National Convention on Statistics (NCS), EDSA Shangri-La Hotel, Quezon City.
- Rogers, T. B. (1995). *The psychological testing enterprise: An introduction*. Brooks/Cole.
- Science Education Institute, Department of Science and Technology & Philippine Council of Mathematics Teacher Education, (2011). *Mathematics framework for Philippine basic education*. Manila: Author.
- UNESCO. (2012). *Challenges in basic mathematics education*.
<http://unesdoc.unesco.org/images/0019/001917/191776e.pdf>
- Urbina, S. (2014). *Essentials of psychological testing*. John Wiley & Sons, Inc.
- Varma, S. (n.d.). *Preliminary item statistics using point-biserial correlation and p-values*.
<https://jcesom.marshall.edu/media/24104/Item-Stats-Point-Biserial.pdf>
- Wiersma, W., & Jurs, S. (1990). *Educational measurement and testing*. Allyn and Bacon.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 12 of 16

Appendices

Table 1. Distribution of Items according to their Difficulty and Discrimination Indices

		Discrimination Index			
		<i>Excellent</i> (.40 & above)	<i>Good</i> (.30 - .39)	<i>Fair</i> (.20 - .29)	<i>Poor</i> (.19 and below)
Difficulty Index	<i>Desirable</i> (.40 - .70)	15, 92	1, 5, 21, 25, 43, 58, 60, 72, 87, 95, 129	3, 36, 45	128
	<i>Difficult</i> (.15 - .39)	2, 103	10, 11, 24, 28, 71, 88, 120	4, 27, 32, 33, 38, 57, 63, 64, 75, 77, 78, 91, 97, 98, 100, 104, 105, 106, 114, 141, 142	6, 7, 8, 9, 12, 13, 16, 17, 18, 19, 20, 22, 23, 26, 29, 30, 31, 34, 35, 37, 39, 40, 41, 42, 47, 49, 50, 51, 52, 53, 55, 56, 59, 61, 62, 65, 66, 67, 68, 69, 70, 74, 76, 79, 80, 81, 82, 83, 84, 85, 86, 89, 90, 93, 94, 96, 99, 101, 102, 107, 108, 109, 110, 111, 112, 113, 115, 116, 117, 118, 119, 121, 122, 123, 124, 125, 126, 127, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 143, 144, 145, 146, 147 148, 149, 150
	<i>Easy</i> (.71 - .85)				
	<i>Very Difficult</i> (.01 - .14)				14, 44, 46, 48, 54, 73
	<i>Very Easy</i> (.86 – 1.0)				

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 13 of 16

Table 2. Final Items Included in the Test w/ their Difficulty & Discrimination Indices

Item No.	Difficulty Index	Interpretation	Discrimination Index	Interpretation
1	0.65	desirable	0.32	good
2	0.34	difficult	0.52	excellent
3	0.65	desirable	0.24	fair
4	0.25	difficult	0.26	fair
5	0.45	desirable	0.34	good
10	0.40	difficult	0.37	good
11	0.36	difficult	0.32	good
15	0.46	desirable	0.42	excellent
21	0.53	desirable	0.37	good
24	0.31	difficult	0.32	good
25	0.49	desirable	0.34	good
27	0.30	difficult	0.20	fair
28	0.39	difficult	0.32	good
32	0.29	difficult	0.25	fair
33	0.29	difficult	0.25	fair
36	0.49	desirable	0.27	fair
38	0.32	difficult	0.20	fair
43	0.48	desirable	0.34	good
45	0.40	desirable	0.30	fair
57	0.38	difficult	0.24	fair
58	0.41	desirable	0.31	good
60	0.45	desirable	0.34	good
63	0.30	difficult	0.20	fair
64	0.29	difficult	0.25	fair
71	0.29	difficult	0.32	good
72	0.48	desirable	0.33	good
75	0.34	difficult	0.20	fair
77	0.34	difficult	0.24	fair
78	0.38	difficult	0.27	fair
87	0.59	desirable	0.37	good
88	0.25	difficult	0.31	good
91	0.26	difficult	0.22	fair
92	0.54	desirable	0.44	excellent
95	0.41	desirable	0.40	good
96	0.33	difficult	0.18	poor
97	0.34	difficult	0.23	fair
98	0.25	difficult	0.23	fair
100	0.31	difficult	0.23	fair
103	0.38	difficult	0.41	excellent
104	0.26	difficult	0.20	fair
105	0.40	difficult	0.20	fair
106	0.30	difficult	0.20	fair
114	0.26	difficult	0.22	fair
118	0.25	difficult	0.19	poor
120	0.27	difficult	0.33	good
123	0.31	difficult	0.19	poor
128	0.43	desirable	0.18	poor
129	0.41	desirable	0.31	good
141	0.25	difficult	0.21	fair
142	0.29	difficult	0.29	fair

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 14 of 16

Table 3. Truncated Results of the Option Analysis

Item No.	Key	Groups	Option A	Option B	Option C	Option D	Blank	Total
1	B	UG	7.69%	81.32%	4.40%	5.49%	1.10%	100%
		LG	18.68%	9.45%	14.29%	16.48%	1.10%	
2	D	UG	20.88%	14.29%	5.49%	59.34%	0.00%	100%
		LG	36.26%	39.56%	15.38%	7.69%	1.10%	
3	C	UG	7.69%	8.79%	76.92%	5.49%	1.10%	100%
		LG	17.58%	23.08%	52.75%	5.49%	1.10%	

Table 4. Correlation Between the Achievement Test Scores and Grades of Students in Senior High School Core Mathematics Subjects

Variables	Grades in Senior High School Core Mathematics Subjects	
	<i>p</i>	<i>p</i> -value
Scores in the 150-item Achievement Test	.437**	.000
Scores in the final 50-item Achievement Test	.504**	.000

**Correlation is significant at the 0.01 level (2-tailed)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 15 of 16

Table 5. Factor Loading for Exploratory Factor Analysis with Varimax Rotation of Achievement Test Items

Items	Factor 1 (General Mathematics)	Factor 2 (Statistics & Probability)
Item 25	.506	
Item 2	.460	.302
Item 10	.450	
Item 11	.395	
Item 21	.368	
Item 33	.355	
Item 4	.346	.104
Item 87	.339	.165
Item 36	.325	
Item 103	.309	.178
Item 95	.304	.232
Item 3	.290	.110
Item 24	.289	
Item 15	.281	.270
Item 77	.276	.233
Item 45	.273	
Item 100	.269	
Item 1	.258	.154
Item 5	.228	.205
Item 71	.224	
Item 106	.218	
Item 91	.210	
Item 129	.203	
Item 28	.193	
Item 141	.192	
Item 64	.190	
Item 32	.177	.109
Item 114	.168	
Item 118		
Item 58		.407
Item 98		.396
Item 60	.147	.387
Item 63		.373
Item 92	.257	.340
Item 43	.158	.331
Item 72		.304
Item 142	.126	.293
Item 88		.271
Item 38		.268
Item 57		.254
Item 105	-.106	.251
Item 128	.166	.219
Item 123	.107	.195
Item 120	.114	.186
Item 78	.149	.177
Item 27	.106	.150
Item 96		.127
Item 97		.120
Item 104		.115
Item 75		.109

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 16 of 16

Table 6. Results of the Reliability Analysis Performed in the 150-item and 50-item Test

Variables	Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of items
150-item Achievement Test	.624	.602	150
50-item Achievement Test	.778	.775	50

Table 7. Results of the Reliability Analysis Performed in the Items in the Two Factors

Variables	Cronbach's Alpha	Cronbach's Alpha Based on Standardized Items	N of items
Factor 1	.743	.741	28
Factor 2	.611	.609	22